

FUSELAGE FRAME – AIRCRAFT STRUCTURES 2 - MECN 4027

CONTEXT & GEOMETRY SETUP

An experimental fuselage composed of n sections was analyzed to study the effect of frame geometry on stress distribution. MATLAB (CEM) and Patran/Nastran (FEM) were used, with FEM results validating the analytical model.

The geometry was generated using trigonometry and point transformations. The outer arc was defined via the cosine rule and rotated using a direct cosine matrix, while the inner geometry was formed by offsetting the outer arc using the specified distances h_1 and h_2 .

EQUATIONS AND ANALYTICAL PROCEEDURE

The first step involved determining a moment distribution due to the loads acting about the structure after it has been cut.

$$M = Ma + Na\Delta y + Va\Delta x + Mp; (Mp = 0 \text{ for Question 1}) \quad (A)$$

Then complimentary energy would be used to solve for the internal loads inside the structure, the derivative of the energy with respect to the load would be set equal to the given mismatch displacements for the first question and zero for the second question

$$\frac{\partial C}{\partial F} = \int_0^s \frac{M}{EI} \cdot \frac{\partial M}{\partial F} ds; \text{ Where } F = [Ma, Na, Va] \quad (B)$$

$$\frac{\partial C}{\partial F} = [\delta_\theta, \delta_x, \delta_y] \text{ (For question 1)}; \frac{\partial C}{\partial F} = 0 \text{ (For question 2)} \quad (C)$$

The axial loads about the structure were estimated at each angular position through a co-ordinate transformation.

$$N(\theta) = Na \cos(\theta) - Va \cos(\theta) + Fpy \sin(\theta) - Fpx \cos(\theta); \quad (D) \\ (Fpy = Fpx = 0 \text{ for Question 1})$$

The stress distribution about the circumference of the structure, would then be estimated. The axial strain energy contribution was also included so as to better improve the accuracy of the results.

$$\sigma = \frac{Mh}{I} + \frac{N(\theta)}{A} \quad (F)$$

After solving for the internal redundant loads, the displacements were then obtained by imposing dummy loads about the structure. A new moment distribution due to the dummy loads was obtained.

$$M = Ma + Na\Delta y_2 + Va\Delta x_2 + Mp + P_v\Delta x_2 + P_h\Delta y_2 + M_\beta \quad (G)$$

$$L = [M_\beta, P_v, P_h]; \frac{\partial C}{\partial L} = [\beta, \Delta h, \Delta v] \quad (H)$$

$$\frac{\partial C}{\partial L} = \int_{s_{ref2}}^s \frac{Ma + Na\Delta y_2 + Va\Delta x_2 + Mp + P_v\Delta x_2 + P_h\Delta y_2 + M_\beta}{EI} \cdot \frac{\partial M}{\partial L} ds \quad (I)$$

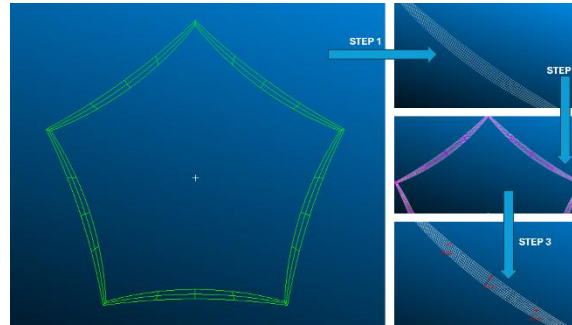
Where :

$$\Delta x = [(x_1 - 0), (x_2 - 0), \dots, (x_n - 0)] \quad (J) \\ \Delta y = [(y_1 - y_{ref}), (y_2 - y_{ref}), \dots, (y_n - y_{ref})]$$

$$\Delta x_2 = [(x_1 - x_{ref2}), (x_2 - x_{ref2}), \dots, (x_n - x_{refn})] \\ \Delta y_2 = [(y_1 - y_{ref2}), (y_2 - y_{ref2}), \dots, (y_n - y_{ref2})]$$

Mp : Moment distribution due to pressure load

FLOW DIAGRAM FEM



FEM FORMULATION

1. Geometry & Meshing

Create geometry in Autodesk Inventor, and section every 18 Degrees and import to Patran

STEP 1, 2 and 3

Create material and assign properties to frame and mesh. Delete duplicate nodes via equivalence. Constrain and solve

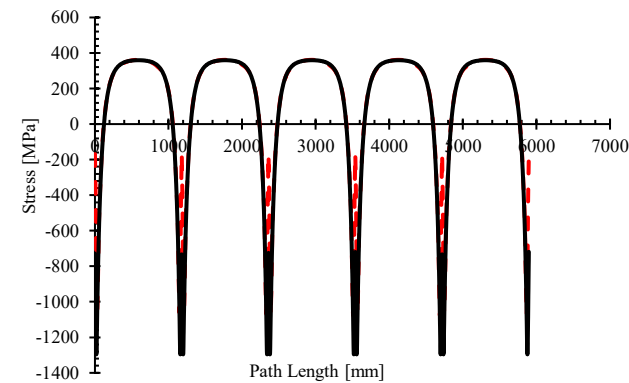
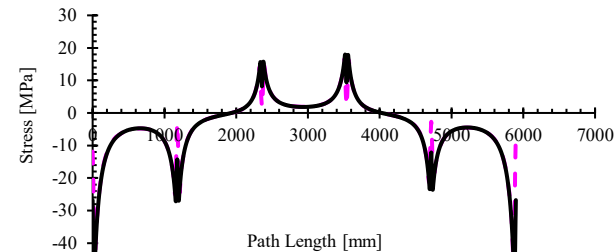
CONSTRAINTS – QUESTION 1

Constraints : Leave duplicate nodes at the top of the frame. Constrain the all nodes at the top left side of the frame in the x-direction and one node at the top of the frame in the y-direction. And All nodes in the z-direction. Then apply a RBE and the mismatch displacements to the nodes on the top right frame

CONSTRAINTS – QUESTION 2

Use half the structure, and bound all nodes at the top and bottom half in the x-direction, one node at the top in the y-direction and all nodes in the z-direction. The apply an edge pressure of -15 bar at outer edges

RESULTS CEM VS FEM



STRUCTURE PROPERTIES → E = 73 GPa; $\nu = 0.3$; Thickness = 10 mm; Frame Spacing = 300 mm

DISCUSSION

RESULTS

CEM-FEM Correlation The correlation coefficient between the CEM and FEM stress distributions was calculated as **0.9863** (Question-1) and **0.9871** (Question-2). These near-unity values explain why the graphs overlay almost perfectly when plotted on the same axes, with the only visible differences occurring at the corners.

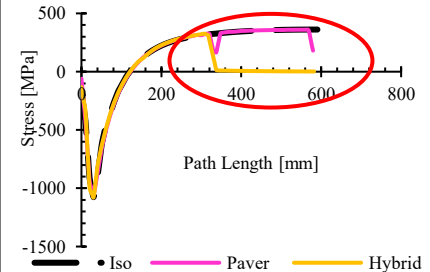
Stress Peaks at Corners (FEM) Sharp corners in the FEM mesh create stress concentrations. The solver drives the stress toward the theoretical infinite value at these cusps. Increasing the local mesh density only sharpens the peaks and pushes the stress values higher.

FEA ANALYSIS AND SELECTIONS

Mesh Choice Iso-mesh Quad 4 elements were selected. They provide uniform, high-resolution stress capture on curved 2D surfaces, unlike triangular elements which create artificial stress concentrations.

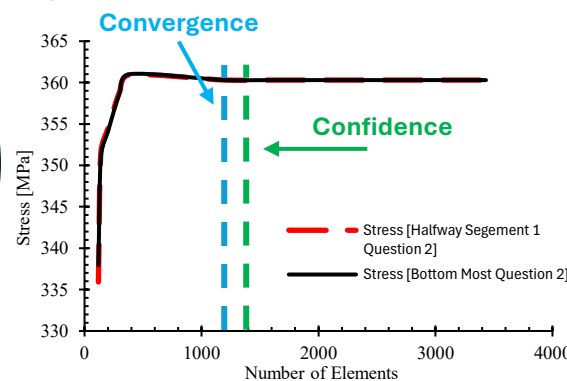
Convergence Stress-based convergence was used. Stresses converge at ~1,200 elements. Displacements converge at ~150 elements. A displacement-based approach (150 elements) would give inaccurate low-resolution stresses, which are critical in real engineering applications

MESH PERFORMANCE

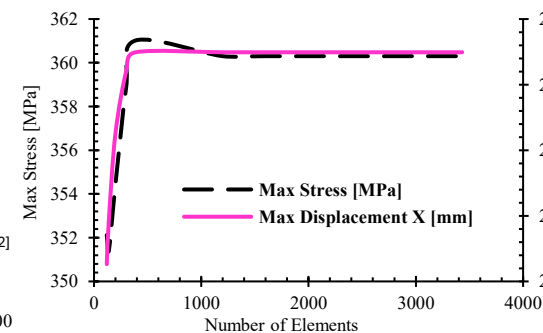


Paver and hybrid meshes showed convergence similar to iso-meshes, with accurate stress distributions up to 300 mm. Beyond this, distortion occurs due to non-uniform triangular and trapezoidal elements, reducing accuracy.

QUESTION 2 – STRESS CONVERGENCE



STRESS VS DISPLACEMENT COMPARISON



DIAGRAMS USED IN CEM FORMULATION

